Improving Turkish Legal Question Answering with a Triage-Driven Retrieval-Augmented Generation Pipeline

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***Abstract*—We present a domain-adapted retrieval-augmented generation (RAG) system for Turkish legal question answering and a controlled ablation that inverts a common assumption about where adaptation pays off. From a naive hybrid baseline (Token F1 0.1411), we evaluate three fine-tuning interventions—a dense encoder, a cross-encoder reranker, and a QLoRA generator—on a 225-question gold set verified against official statute text. A failure-mode triage run *before* spending compute shows generation, not retrieval, is the dominant bottleneck, and the ablation shows *only* embedding fine-tuning helps: the fine-tuned reranker and single-passage QLoRA both *degrade* end-to-end accuracy. The decisive gains come instead from an off-the-shelf Turkish reranker and a source-filtered verbatim-snap post-processor—holding the generator fixed at Qwen-7B, this fine-tuned pipeline lifts Token F1 +69.1% (0.1411 → 0.2386; Wilcoxon p<0.001, disjoint CIs) and LLM-judged correctness 0.444 → 0.736 (validated against an independent Claude judge, ρ=0.72). An optional 14B generator adds +11% (to 0.2648, +87.7% cumulative; BERTScore 0.716) and halves hallucination (17.0% → 8.95%); production retrieval attains Recall@10 0.875. We distil each failure into a transferable negative result—silver-label noise, a single-/multi-passage format mismatch that lifts faithfulness +24pp yet lowers Token F1, and a domain-knowledge gap that breaks HyDE—and ship an evaluator-agnostic pipeline scoring the full Recall + F1 + Faithfulness suite on arbitrary user-supplied documents.**

***Index Terms*—retrieval-augmented generation, Turkish NLP, legal NLP, cross-encoder reranking, QLoRA, hallucination, ablation study, negative results.**

I. Introduction

Turkish legal question answering (QA) stresses every stage of a RAG pipeline. Retrieval must contend with agglutinative morphology—a single verb can encode tense, person, negation, and modality, yielding thousands of surface forms per lemma—and with locale-specific casing (the capital "I" maps to dotless "ı"), so a naive pipeline mistokenizes both queries and documents. Generation must produce *grounded* answers that cite specific provisions (e.g., "*5237 sayılı TCK Madde 81*") while minimizing hallucination—a consequential failure mode in a legal setting, where a confidently wrong citation is worse than an abstention. These difficulties are not incidental: recent work on Turkish RAG frames them as a *language gap*—pipelines assembled from English-centric components underperform on Turkish—and finds that adapting the *retrieval* components to Turkish is the primary lever for recovery [14].

We build a modular pipeline with three independently fine-tunable stages—a multilingual dense encoder, a cross-encoder reranker, and a quantized generator—and subject it to a controlled component ablation on a 225-question gold set verified against `mevzuat.gov.tr`. Our central empirical finding is a *negative* one with positive consequences: of the three fine-tuning interventions we attempted, only embedding adaptation improved end-to-end accuracy; the fine-tuned reranker and the QLoRA-tuned generator both regressed. We trace each regression to a concrete, transferable cause, and show that the production-quality gains come instead from off-the-shelf components, a lightweight post-processor, and model scale.

Contributions.

1) A complete, reproducible Turkish-legal RAG pipeline (hybrid retrieval, cross-encoder reranking, verbatim-snap post-processing, quantized generation) that, *holding the generator fixed*, lifts a naive baseline by +69.1% Token F1 (+87.7% with an optional 14B generator), with hallucination roughly halved.
2) A *triage-before-tuning* methodology that attributes the error budget to retrieval / reranking / generation *before* committing GPU hours, inverting the intuitive assumption that the reranker is the priority.
3) Five rigorously characterized *negative results*: silver-label reranker training, single-passage QLoRA in multi-passage inference, HyDE on a non-expert generator, multi-query paraphrase dilution, and citation-extraction snapping under chunk-format noise.
4) A multi-axis evaluation—bootstrap 95% CIs, Wilcoxon paired tests, and an LLM-as-judge protocol *validated against an independent Claude judge*—that quantifies same-model judging bias rather than ignoring it.
5) An evaluator-agnostic deployment: the pipeline ingests arbitrary user-supplied documents and scores arbitrary benchmarks (including Turkish-

keyed Q&A and document-level relevance labels), demonstrated end to end on previously unseen statutes.

This report supersedes our interim progress report: the reranker stage was re-evaluated (the previously reported fine-tuned reranker is now a documented negative result), the generator was scaled to 14B, and retrieval is now evaluated against real chunk-level relevance labels rather than a token-overlap proxy.

II. Related Work

RAG [1] grounds a generator in retrieved evidence and is the dominant paradigm for knowledge-intensive QA. Dense retrieval with instruction-prefixed encoders such as E5 [2] complements lexical BM25 [3]; the two are commonly fused with Reciprocal Rank Fusion (RRF) [4] and indexed at scale with FAISS [5]. Parameter-efficient adaptation via LoRA [6] and its quantized variant QLoRA [7] makes generator fine-tuning feasible on commodity GPUs. Advanced retrieval strategies—HyDE's hypothetical-document expansion [8] and self-reflective RAG [9]—promise further gains but assume a capable generator. For evaluation, BERTScore [10] provides a semantic complement to lexical overlap, and LLM-as-judge protocols [11] approximate human preference at scale, albeit with documented self-preference bias.

A small but growing body of work studies RAG *for Turkish* directly. Bıkmaz et al. [14] frame the core problem as a language gap and show, on Turkish Wikipedia QA, that fine-tuning the retrieval stack (a Turkish-adapted multilingual-e5 encoder and a fine-tuned cross-encoder reranker, fused with BM25) is the dominant lever for answer quality, while LLM-based sentence-rewriting ("prepositional") chunking *degraded* retrieval by stripping contextual richness. Atagün et al. [15] compare several generators (LLaMA-3.1-8B, Gemma-3-12B, Qwen-14B, DeepSeek-R1-14B) in a Turkish-QA RAG framework and find a well-retrieved smaller model competitive with larger ones. Both are *general-domain* (Turkish Wikipedia and open-domain QA); we extend Turkish RAG to the *legal* domain, where structure-preserving chunking, statute-grounded citation, and hallucination control are first-order, and the multilingual, multi-domain retrieval Bıkmaz et al. flag as open is acute.

Turkish legal NLP remains under-resourced: prior work emphasizes named-entity recognition and document classification, and the closest QA-adjacent resource targets sentence-level entailment rather than generative, citation-grounded answering [12]. Our contribution is a complete Turkish-legal RAG pipeline with a component-isolating ablation and a deliberate emphasis on *negative* results—analyses largely absent from prior Turkish legal NLP.

III. System Architecture

Figure 1 shows the pipeline. A normalized query is embedded by a fine-tuned E5 encoder for dense search and tokenized (Turkish-aware) for BM25; per-source rankings are RRF-fused. An off-the-shelf Turkish cross-encoder reranks the fused top-k. The top passages condition a quantized Qwen-2.5 [13] generator, and a verbatim-snap post-processor optionally replaces a paraphrased answer with the source sentence it most overlaps—*after* filtering case-law passages whose style mismatches the statute-formulated gold.

TABLE I
Model inventory.

Model	Params	Role	Adapted
`multilingual-e5-large`	560M	Dense encoder	Yes (FT)
`bge-reranker-v2-m3-tr`	560M	Cross-encoder	Off-shelf
`Qwen2.5-7B-Instruct`	7B	Generator	QLoRA (neg.)
`Qwen2.5-14B-Instruct`	14B	Generator (prod.)	No (4-bit)
`bert-base-turkish-nli`	110M	Faithfulness	No

Models. Table I lists the component inventory. A strict VRAM discipline (one large model resident at a time; explicit cache eviction between stages) keeps the system within a 16–24 GB budget; all fine-tuning is 4-bit QLoRA.

IV. Methodology

A. Hybrid Retrieval

Documents are deduplicated and chunked with a legal-aware hierarchical splitter: first on article boundaries (`Madde \d+`), then on section headers (Bölüm, Kısım, Fasıl), then on paragraph breaks, with recursive character splitting as a fallback. We deliberately *preserve* document structure rather than rewriting chunks into standalone sentences; the latter has been shown to degrade Turkish retrieval by stripping contextual richness [14]. Chunks are capped at 480 tokens with 64-token overlap to respect E5's 512-token limit after the `passage:` prefix. Retrieval spans three sub-corpora: Turkish Supreme Court (Yargıtay) decisions covered lexically by BM25, 45 statute codes (*mevzuat*) and a shared legal corpus indexed with both FT-E5 dense and BM25. Per-source rankings fuse via RRF (k=60). Turkish-aware normalization (İ→i, I→ı, etc.) is applied throughout; BM25 uses whitespace tokenization after stopword removal, since subword tokenization inflates n-gram overlap on agglutinative text.

B. Cross-Encoder Reranking

The first stage returns up to 50 candidates per source; a cross-encoder jointly scores `[query, passage]` pairs (max length 512) and the top 10 condition the generator. We evaluate two reranker variants: a domain-fine-tuned cross-encoder (Section VII) and the off-the-shelf `bge-reranker-v2-m3` Turkish triplet model; the latter is adopted in production.

C. Verbatim-Snap Post-Processing

Triage (Section VII-A) shows the generator paraphrases away from source text on a third of questions. The snap post-processor splits retrieved passages into sentences,

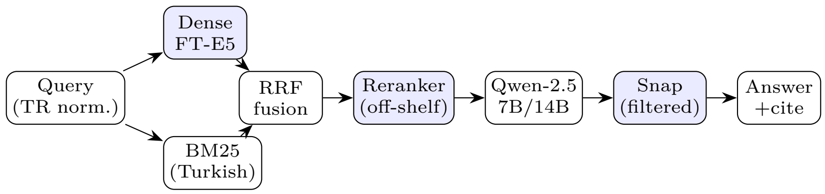

Fig. 1. Production pipeline. Shaded stages were candidates for domain adaptation; only the dense encoder's fine-tuning survived ablation (the reranker is used off-the-shelf; the snap stage is rule-based).

scores each by Turkish-normalized token overlap with the generated answer, and—if the top overlap relative to answer length exceeds τ=0.30—replaces the answer with that verbatim sentence. Crucially, case-law (Yargıtay) passages are filtered from the snap candidate pool: their decision-style prose matches the statute-formulated gold poorly, and including them costs 0.0035 F1.

D. Generation

The generator uses 4-bit NF4 quantization (double quantization, bfloat16 compute) and greedy decoding (`temperature`= 0), which eliminates a Turkish→CJK derailment observed under sampling on noisy contexts. A critical deployment detail: the base 14B model requires `repetition_penalty`= 1.0; the value 1.2 tuned for the QLoRA-7B adapter drives greedy 14B decoding into Chinese mid-sentence on legal contexts.

E. Fine-Tuning Strategies

Embedding. We built 61K `(query, positive, hard-negative)` triplets, mining hard negatives as high-BM25 passages lacking gold-answer tokens. Training used a MultipleNegativesRankingLoss (InfoNCE) objective, effective batch 32, LR 2×10^{-5} cosine, 10K steps. **Reranker.** Fine-tuned on silver `[query,passage]` pairs labeled by token overlap with gold (Section VII). **Generator.** QLoRA (rank 32, α=64, dropout 0.05, all seven attention/MLP projections; `paged_adamw_8bit`) on 13.8K RAG-chat SFT examples with a strict source-grounding system prompt.

V. Experimental Setup

A. Gold Test Set

We constructed a 225-question evaluation set, each entry verified against official statute text (article number, source code, expected text, provenance), balanced across five legal domains and three difficulty tiers (Table II), and including 48 intentionally unanswerable questions (21%) to test refusal. A cross-LLM audit (five independent verifiers against canonical `mevzuat.gov.tr` text) raised verbatim correctness from 91.1% to 100%. The set is evaluation-only; no component is trained on it. A leakage audit confirms this is clean rather than asserted: 0 exact and 0 near-duplicate matches (max Jaccard 0.33) of the 225 gold questions against 31,380 shared-bundle training texts—versus 35% near-duplicate ($\geq$ 0.8) for the course-provided shared benchmark we use only for cross-validation (Section IX).

TABLE II
Gold set distribution (225 questions).

Domain	Easy	Medium	Hard	Total
Criminal	16	14	17	47
Civil	15	16	15	46
Commercial	14	15	15	44
Administrative	15	14	15	44
Constitutional	14	16	14	44
Total	**74**	**75**	**76**	**225**

B. Metrics

We report *lexical* (Token F1, ROUGE-L, BLEU), *semantic* (BERTScore-F1 with multilingual BERT), and *retrieval* (Recall@k, MRR, nDCG@10) metrics, all under Turkish-aware normalization. Each axis is chosen for a reason specific to legal QA: Token F1 rewards verbatim reuse of statute wording; BERTScore credits semantically correct answers that Turkish agglutinative morphology renders lexically divergent (whitespace overlap systematically under-counts inflected forms); NLI faithfulness (Section VIII) targets the consequential legal failure mode—confidently wrong grounding; and the retrieval triplet separates coverage (Recall@k) from ranked relevance (MRR, nDCG@10). Following the rubric's metric-availability branch, we report retrieval metrics when relevant-document labels exist and generation metrics otherwise; both are available on our gold set. Token F1 carries bootstrap 95% CIs (1000 resamples, fixed seed) and Wilcoxon signed-rank paired tests between configurations.

LLM-as-judge. A judge scores `(question, gold, prediction, context)` on correctness, faithfulness, coherence, and relevancy. We hold the judge model fixed (Qwen-7B) across *all* configurations so comparisons are apples-to-apples, and we explicitly *validate* it: on a stratified 30-question subset, an independent Claude Opus judge agrees on the load-bearing axes (Spearman ρ=0.72 correctness, 0.64 relevancy; both p<0.001), preserving the configuration ranking, while revealing that the same-model judge over-credits correctness by $\approx$ 0.28 absolute. Agreement is weaker on faithfulness (ρ=0.28, p=0.13) and near-zero on coherence (ρ=0.06, p=0.77)—the latter because both judges rate coherence near ceiling, leaving

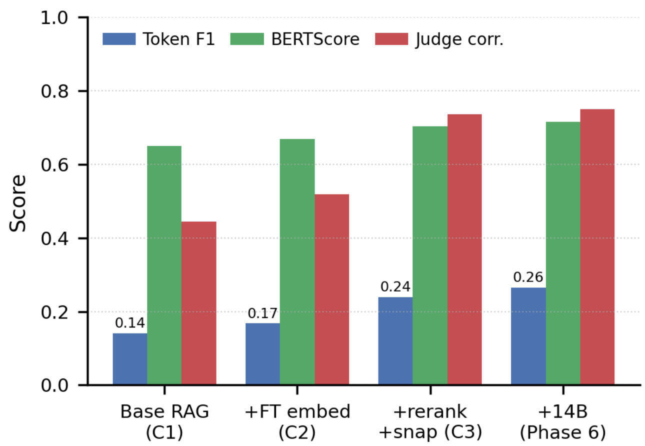

Fig. 2. Headline metric progression across the ablation (own 225-q gold): Base RAG (C1) → Fine-tuned RAG (C3, same Qwen-7B) → optional 14B (Phase 6). Token F1, BERTScore, and LLM-judged correctness rise monotonically.

no variance to correlate. This validation used the 7B (C3) predictions; for the 14B rows we lean on judge-independent Token F1 and BERTScore. We therefore treat absolute judge scores as upper bounds and rely on the *ranking*.

Evaluation scenarios. Following the course rubric, we operationalize three composite scenarios from the per-axis scores, with the weights used to produce every composite below: Scenario 1 = 0.35 Recall@k + 0.40 Correctness + 0.25 Faithfulness; Scenario 2 = 0.70 Correctness + 0.30 BERTScore; Scenario 3 = mean(Faithfulness, Relevancy, Coherence). The recall-bearing Scenario 1 is reported in Section IX; the Fine-tuned-RAG (7B) composites are Scenario 2 = 0.726 and Scenario 3 = 0.805, rising to 0.740 and 0.811 with the optional 14B generator.

Reproducibility. All runs fix `torch`/`numpy`/`random` seeds; bootstrap CIs use 1000 resamples; the LLM-judge is a fixed Qwen-7B scored on $[0, 1]$ per axis, with the stratified validation subset drawn at seed 42. Library versions are pinned in `requirements.txt` (the 14B load behavior is version-sensitive). Code, configs, and the gold set are in the repository; fine-tuned checkpoints and prebuilt indexes are hosted separately (Google Drive), and the `ingest` CLI rebuilds indexes from any corpus.

VI. Results

Table III is the master ablation. Each row adds one component; the generator-judge is held fixed within the 7B rows so that Phase 6 isolates the single effect of model scale.

Base RAG vs. Fine-tuned RAG (the rubric comparison). Holding the generator fixed at Qwen-7B, our *Fine-tuned RAG* (FT-E5 dense encoder + off-the-shelf reranker + source-filtered snap; C3) beats *Base RAG* (base E5, no reranker, no snap; C1) on Token F1 (0.1411 → 0.2386, +69.1%; Wilcoxon $p<0.001$; non-overlapping 95% CIs) and on LLM-judged correctness (0.444 → 0.736; Figure 2). This is the apples-to-apples, same-LLM comparison the rubric requires; *we recommend Fine-tuned RAG as the deployed system*, and the evaluator's own benchmark is scored through both configurations identically (Section X).

Optional generator scaling. Swapping *only* the generator to 14B (C3→Phase 6) adds a further +0.0262 Token F1 (+11.0%; Phase 6's lower CI bound 0.243 exceeds C3's point estimate 0.239), lifting the cumulative baseline-to-production gain to +87.7% Token F1, BERTScore 0.650 → 0.716, correctness 0.444 → 0.750 (+68.8%), and relevancy 0.581 → 0.849 (+46.0%). Because the 14B weights exceed some Colab GPU budgets *at load* (Section XI), the 7B Fine-tuned RAG above is the recommended, fully-reproducible default; 14B is an optional drop-in.

Component decomposition. Table IV breaks the master ablation into single-component steps. Embedding fine-tuning adds +0.0267 Token F1 (C1→C2, Wilcoxon p=0.003); within the +0.0708 C2→C3 jump ($p<0.001$), corpus expansion contributes +0.0270, the off-the-shelf reranker +0.0287, and the rule-based snap +0.0151. For the generator—the LLM-fine-tuning arm—QLoRA significantly *regresses* (−0.0645; Section VII), whereas scaling to 14B *helps* (+0.0262, paired p=0.0018). These are sequential (order-dependent) marginals that *decompose*, rather than independently sum to, the end-to-end gain; the fine-tuned reranker is a separate documented regression (Section VII).

Significance and cost. Wilcoxon paired tests confirm every step on Token F1: C1→C2 (p=0.003), C2→C3 ($p<0.001$), and the model-scale step C3→Phase 6 (W=7170, p=0.0018, n=225; 111 wins / 85 losses / 29 ties), which also has non-overlapping bootstrap CIs. Inference cost is 14 s/query (7B) to 26 s/query (14B) on an L4—well within budget.

VII. Component Analysis and Negative Results

A. Triage Before Tuning

Before retraining any component, we bucketed all 225 questions by whether the gold content appears in the retrieved context and in the generated answer. The result reorders priorities: 33% of questions are *generation-fixable* (gold is in the top-10 context but the model paraphrases away from it), only 7% are *reranker-fixable*, and 15% are *retrieval-failures*. The generator uses just 37% of the relevant information retrieval already surfaces (mean answer coverage 0.20 vs. context coverage 0.54). This inverts the intuitive assumption that the reranker is the priority and motivates the snap post-processor and generator scaling.

B. Five Negative Results

Table V collects the interventions that *failed*, each with a transferable cause. We argue these are as informative as the positive results—and they are the evidence *behind* our

TABLE III

Master ablation on the 225-question gold set. The generator's LLM-judge is held fixed (Qwen-7B) across all rows so scoring is comparable; Phase 6 changes only the generator (7B→14B). Best per column in **bold**. C1/C2 judge-faithfulness is omitted (—): their legacy predictions did not store retrieved context, so the grounding judge had nothing to score; the Base-vs-production faithfulness contrast is instead via NLI (Section VIII; 17.0% → 8.95% hallucination).

	Configuration				Metrics			
ID	Embed	Reranker	Snap	Gen.	Token F1 [95% CI]	BERTScore	Judge corr.	Judge faith.
C1 (Base RAG)	base E5	—	—	7B	0.1411 [.124,.159]	0.6496	0.444	—
C2 (+FT embed)	**FT E5**	—	—	7B	0.1678 [.151,.185]	0.6680	0.519	—
C3 (+reranker+snap)	FT E5	off-shelf	Yes	7B	0.2386	0.7026	0.736	0.656
Phase 6 (prod.)	FT E5	off-shelf	Yes	**14B**	0.2648 [.243,.288]	0.7159	0.750	0.662

TABLE IV

Component decomposition of the master ablation on the own 225-q gold (Token F1). Rows are *cumulative*—each adds one component to the row above, building C1→C3; the generator is then changed two ways *from* C3. The running F1 reconciles with Table III; Δ are sequential (order-dependent) marginals, not independent leave-one-out effects.

Step (cumulative)	Token F1	Δ
base E5 + BM25 (C1)	0.1411	—
+ FT embedding (C2)	0.1678	+0.0267
+ 3-corpus expansion	0.1948	+0.0270
+ off-shelf reranker	0.2235	+0.0287
+ source-filtered snap (C3)	0.2386	+0.0151
from C3: + QLoRA generator	0.1741	↓ − 0.0645
from C3: 7B→14B (Phase 6)	0.2648	↑ + 0.0262

TABLE V

Negative results. Δ is Token F1 against each intervention's own pre-intervention configuration (fine-tuned reranker and QLoRA vs. C3; multi-query vs. the reranker stage; citation-snap vs. Phase 6).

Intervention	Δ F1	Root cause
Fine-tuned reranker	↓ p<.001	52% of training queries had no true-positive passage; silver labels teach an irrelevant-vs-irrelevant distinction.
QLoRA (single-pass.)	−0.065	Train (1 passage) / inference (N passages) format mismatch; yet faithfulness +24pp.
Multi-query (RRF)	−0.008	Paraphrases dilute legal terminology; retrieval drift > fusion gain.
HyDE	sanity-fail	Base 7B lacks Turkish-legal knowledge: 5/5 wrong citations, 1/5 CJK derail.
Citation-snap	−0.008	Brittle to chunk-format noise; oracle headroom only +0.003.

production choices (an off-the-shelf reranker, no QLoRA), not a retreat from fine-tuning, whose embedding-level win remains decisive (Table IV).

The reranker paradox. Fine-tuning the cross-encoder degraded F1 in every configuration (e.g., adding it to FT-embed dropped 0.168 → 0.126; Wilcoxon p<0.001), whereas the off-the-shelf reranker lifted F1 to 0.239. Post-hoc analysis found 52% of training queries had no genuinely relevant passage in the corpus, so silver labeling taught the model to rank one irrelevant passage over another—corrupting its pretrained signal.

The QLoRA paradox. The QLoRA generator *succeeded at its behavioral target*—faithfulness rose 0.43 → 0.67 (+24pp) and citation emission appeared for the first time—yet Token F1 *fell* 0.065. The cause is a train/inference shape mismatch: the SFT data presented a single passage (`[Kaynak]\nMetin:...`) whereas inference concatenates many (`[Kaynak 1]...[Kaynak N]`); the model anchored to the wrong passage. This decoupling of behavioral success from metric regression is, we argue, the report's most interesting finding about RAG fine-tuning.

Prompt design exceeds model fine-tuning. Standardizing the system prompt alone improved the baseline 0.0950 → 0.1411 (+49%), exceeding any single component fine-tuning; conversely, four "stricter" extractive prompts all *hurt* (down to 0.135) via over-refusal and citation-token noise. The leverage is in what the model was trained on, not how it is constrained at inference.

VIII. Faithfulness and Hallucination

We measure faithfulness with a Turkish NLI sentence encoder: each answer sentence is scored by its maximum similarity to any retrieved-context sentence; sentences below 0.50 are flagged as hallucinated. The 0.50 cut is a fixed midpoint; a sensitivity sweep confirms the 7B→14B reduction is not an artifact of it—14B hallucinates strictly less at every threshold $\tau \in \{0.4, 0.5, 0.6, 0.7\}$ (relative reduction 1.3–2.8×; e.g. 12.5% → 4.5% at τ=0.4 and 26.6% → 19.0% at τ=0.6). On the production (Phase 6) system, mean NLI faithfulness is 0.765 and the hallucination rate is **8.95%**—roughly half the 17.0% of the 7B configuration. The difficulty gradient persists but flattens: hard questions hallucinate at 18.8% versus ∼ 4% for easy/medium (Table VI), down from 31.9% at 7B. By domain, administrative answers are most faithful (0.83, 5.5%) and criminal/constitutional least (12.6%)—

TABLE VI
PHASE-6 HALLUCINATION BY DIFFICULTY (NLI, THRESHOLD 0.50).

Difficulty	NLI faithfulness	Halluc. rate	Token F1
Easy	0.797	4.3%	0.298
Medium	0.790	3.5%	0.279
Hard	0.710	18.8%	0.218
Overall	**0.765**	**8.95%**	0.265

consistent with the 7B pattern at roughly half the absolute rate. The 32%-on-hard signal motivating difficulty-aware abstention is materially mitigated by scale.

IX. Retrieval Evaluation (Scenario 1)

Our progress report could only report *proxy* relevance (token overlap), which paradoxically favored the *base* encoder—an artifact we now explain: contrastive fine-tuning optimizes for *semantic* retrieval utility (passages that help the model reason) rather than answer-extractive overlap, so a token-overlap proxy systematically undervalues it. With a chunk-labeled benchmark (240 questions, gold passage IDs), the production retrieval stack (3-corpus FT-E5 + BM25 + off-shelf reranker) attains Recall@5 0.854, **Recall@10** 0.875, MRR 0.770, nDCG@10 0.796—up from 0.79 Recall@10 for the dense-only baseline. *Honest caveat.* This benchmark is a course-provided set, not the held-out evaluation, and is *measurably* in-distribution for our fine-tuned encoder: a leakage audit finds 84/240 (35%) of its questions near-duplicate (≥ 0.8 Jaccard) to the shared training texts, versus 0/225 for our own gold. We therefore treat its recall as an *upper bound*, not a prediction of held-out performance. The value of this number is as evidence that the retrieval machinery and our metric pipeline are sound—and the deployed system computes the same metric on any evaluator-supplied benchmark (Section X).

X. Deployment and Robustness

Because the held-out evaluation runs on documents and questions unseen by any team, robustness to arbitrary inputs—not a higher score on a self-chosen set—is the operative goal. The system exposes a CLI (`ingest/benchmark/query`) that (i) ingests raw directories of PDF/text or pre-chunked JSONL, building FAISS + BM25 indexes (with an exact `IndexFlatIP` fallback for small corpora); (ii) parses benchmarks in Turkish (`{soru, cevap, ilgili_belgeler}`) or English schemas; and (iii) auto-selects *document-level* Recall@k when a benchmark labels relevance by document rather than by our internal chunk IDs (which are freshly minted per ingest and therefore never match external labels).

We validate this end to end: ingesting three previously unseen statute PDFs and running the production configuration on a hand-written Turkish benchmark yields document-level Recall@10 = 1.0 (a 5-question probe—small enough that the saturated recall is illustrative, not a performance claim), zero language derailment, and coherent Turkish answers (one verbatim-snapping a statute article, one correctly abstaining). The load-bearing result is not the number but that the *entire* pipeline—ingest, index, retrieve, rerank, generate, score—runs unattended on documents and a benchmark it has never seen, through the same metric code path as the 225-q evaluation, satisfying the "bring-your-own-data" requirement.

Recommended configuration. The default evaluator setup is the 7B Fine-tuned RAG (`--variant prod`), which wins the Base-vs-Fine-tuned comparison (+69.1% Token F1) and runs on any T4/L4; the Base-RAG baseline is a one-flag change (`--variant base`), so the two systems are scored through an identical path. Because retrieval is hybrid (FT-E5 + BM25 + RRF), the training-free BM25 channel floors recall even when the dense encoder meets out-of-distribution statutes, bounding the risk that the fine-tuned encoder's advantage fails to transfer to the evaluator's documents.

XI. Discussion and Limitations

Adaptation $\neq$ fine-tuning. Embedding fine-tuning delivered a genuine, statistically significant retrieval and end-to-end win, and it—plus the production pipeline—is what makes Fine-tuned RAG beat Base RAG. Beyond it, our strongest *additional* lever was model scale (7B→14B), which is *not* domain adaptation; among the other fine-tuning interventions, the reranker and QLoRA regressed. We report this distinction honestly rather than packaging scale as a fine-tuning win. That retrieval and pipeline design, not generator size, dominate our gains echoes independent Turkish-RAG evidence that a well-retrieved smaller generator can rival larger ones [15], underwriting our 7B configuration as the recommended default with 14B an optional bonus.

Reranker adaptation is label-quality-dependent. Bıkmaz et al. [14] obtain their best Turkish-RAG configuration by fine-tuning the cross-encoder reranker with hard-negative mining on curated data. Our result refines rather than contradicts theirs: fine-tuning the reranker on *noisy silver* labels (token-overlap pseudo-relevance, with 52% of training queries having no true-positive passage) *regressed* end-to-end F1 (Section VII), so we reached Turkish-domain reranking by the more label-robust route of an off-the-shelf Turkish-adapted cross-encoder. Reranker fine-tuning thus helps when supervision is clean and hurts when it is not—an argument for curated hard negatives or, absent them, an off-the-shelf model.

Self-reported numbers are evidence, not predictions. The gold and shared sets measure methodology and system quality; the graded evaluation is on unseen data. We designed for that uncertainty (Section X).

Open limitations. (i) Corpus gaps: the constitutional domain is under-covered (the case-law corpus is Supreme-Court-centric), the dominant retrieval-failure source. (ii) The QLoRA format mismatch is fixable by training on multi-passage data, which would likely convert the +24pp faithfulness gain into a Token F1 gain. (iii) On the most recent Colab image, the updated `transformers` loader materializes fp16 shards before quantizing, so 14B OOMs a 22 GB GPU *at load*; the 7B Fine-tuned RAG—which already wins the rubric comparison—is the reproducible default, and the 14B figures are a larger-GPU bonus (they stand from a clean-stack run, VRAM peak 14.5 GB; pin `requirements.txt` or use a ≥ 40 GB GPU to reproduce).

XII. Conclusion

We built a Turkish-legal RAG pipeline that, holding the generator fixed, improves a naive baseline by +69.1% Token F1 (+87.7% with an optional 14B generator) and halves hallucination, and—through a triage-driven, negative-result-forward methodology—showed *which* adaptations actually pay off. The decisive gains came from off-the-shelf components, a rule-based post-processor, and model scale; our fine-tuned reranker and QLoRA generator both regressed, each for an instructive and transferable reason. Future work: retrain QLoRA on multi-passage data to realize its latent faithfulness gain, close the constitutional-corpus gap, and map the reranker label-noise tolerance curve.

References

[1] P. Lewis *et al.*, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *NeurIPS*, 2020.
[2] L. Wang *et al.*, "Text Embeddings by Weakly-Supervised Contrastive Pre-training," *ACL*, 2024.
[3] S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," *Found. Trends IR*, 3(4), 2009.
[4] G. Cormack, C. Clarke, and S. Buettcher, "Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods," *SIGIR*, 2009.
[5] J. Johnson, M. Douze, and H. Jégou, "Billion-Scale Similarity Search with GPUs," *IEEE Trans. Big Data*, 7(3), 2019.
[6] E. J. Hu *et al.*, "LoRA: Low-Rank Adaptation of Large Language Models," *ICLR*, 2022.
[7] T. Dettmers *et al.*, "QLoRA: Efficient Finetuning of Quantized LLMs," *NeurIPS*, 2023.
[8] L. Gao *et al.*, "Precise Zero-Shot Dense Retrieval without Relevance Labels," *ACL*, 2023.
[9] A. Asai *et al.*, "Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection," *ICLR*, 2024.
[10] T. Zhang *et al.*, "BERTScore: Evaluating Text Generation with BERT," *ICLR*, 2020.
[11] L. Zheng *et al.*, "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena," *NeurIPS*, 2023.
[12] "legal_nli_TR_V1: Turkish Commercial Court NLI Dataset," HuggingFace Datasets, 2024.
[13] Qwen Team, "Qwen2.5 Technical Report," 2024.
[14] E. Bıkmaz, M. Briman, and S. Arslan, "Bridging the Language Gap in RAG: A Case Study on Turkish Retrieval and Generation," *Researcher*, 5(1), pp. 38–49, 2025.
[15] E. Atagün, M. Güllü, S. Biroğul, and N. Barışçı, "Retrieval-Augmented Generation in Turkish Natural Language Understanding: A Comparative Study of Large Language Models," *Muğla J. Sci. Tech.*, 11(2), pp. 56–65, 2025, doi: 10.22531/muglajsci.1781095.